REPORT ON

Implementation of DDL & DML Operations in Databricks

Team Details

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1. Objective

To implement and demonstrate all major **DDL (Data Definition Language)** and **DML (Data Manipulation Language)** operations using **Databricks SQL** on a real-world dataset.

2. Use Case

We selected a **Weather Data Analysis System** where:

- Weather data is stored and managed
- Data is cleaned and transformed
- Insights like average temperature and rainfall category are derived
- Data updates and corrections are handled efficiently

3. Dataset Description

- **Dataset Name:** weather_data.csv
- **Type:** Structured CSV
- **Domain:** Environmental / Weather Monitoring

Attributes:

Column	Description
district	District name
mandal	Region within district
weather_date	Date of observation
rain	Rainfall (mm)
min_temp	Minimum temperature
max_temp	Maximum temperature
min_humidity	Minimum humidity
max_humidity	Maximum humidity
min_wind_speed	Minimum wind speed
max_wind_speed	Maximum wind speed

4. Tools & Technologies

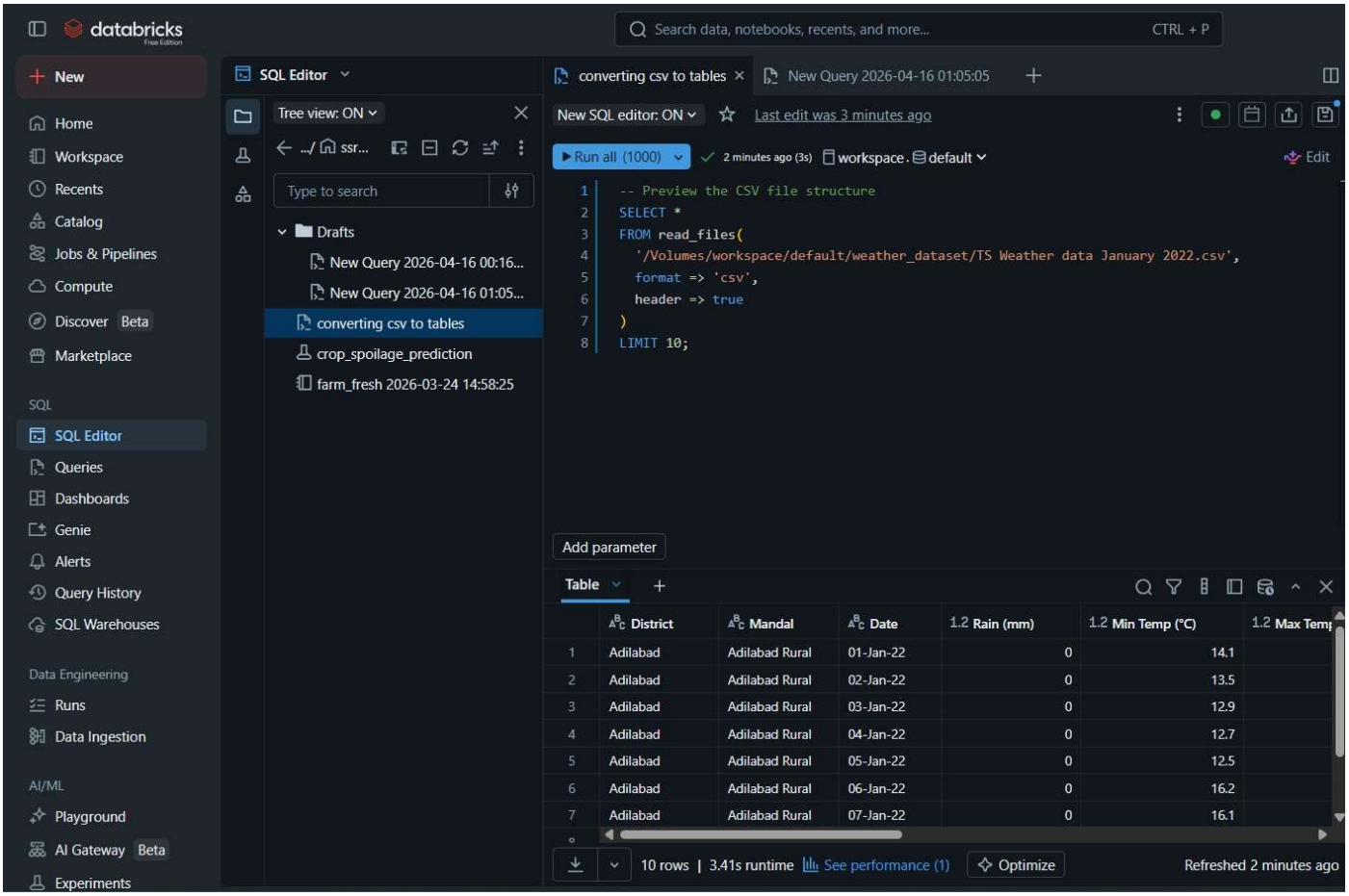
- **Platform:** Databricks
- **Language:** SQL (Databricks SQL)
- **Storage:** Delta Lake
- **Interface:** SQL Editor

5. Data Ingestion Process

1. Uploaded weather_data.csv using Databricks UI
2. Selected:

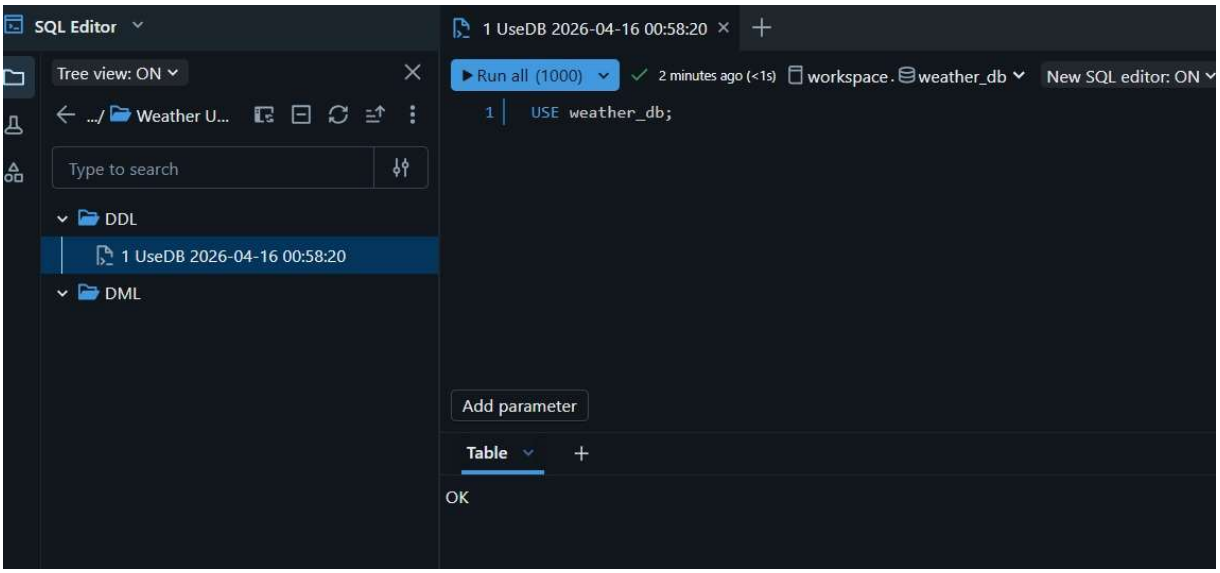
    - Catalog: workspace
    - Schema: weather_db

3. Renamed columns to remove spaces and special characters
4. Created Delta table weather_data

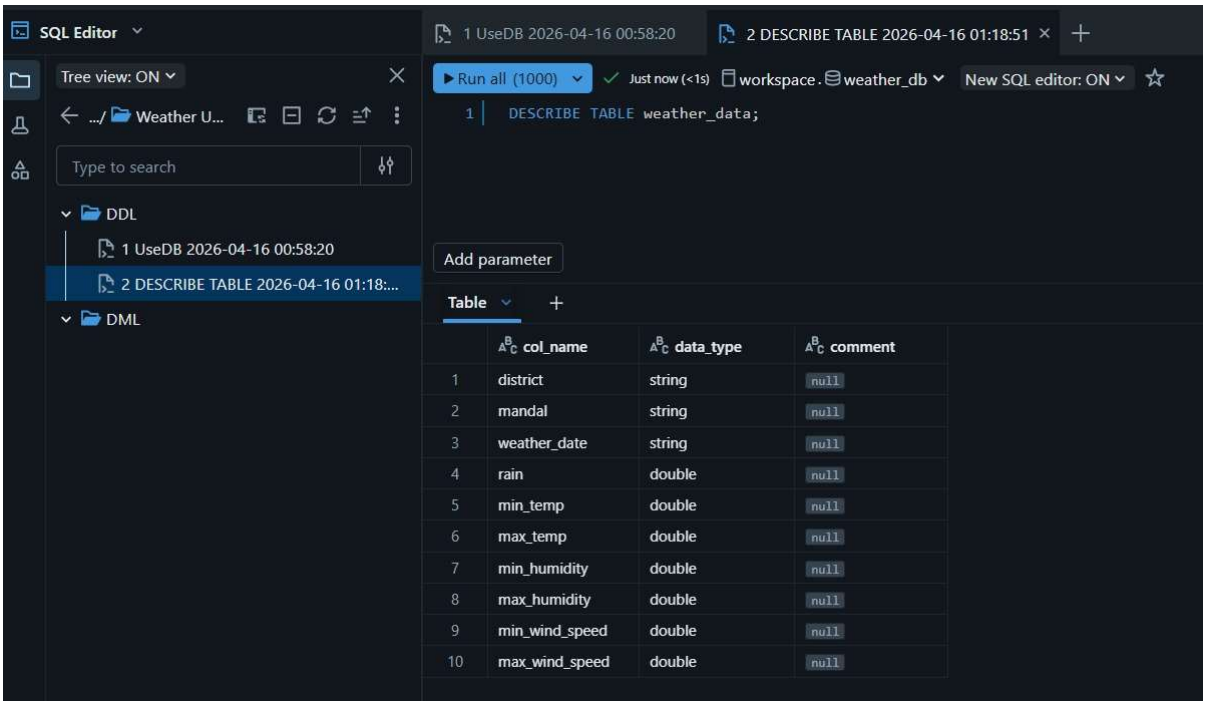


6. DDL Operations

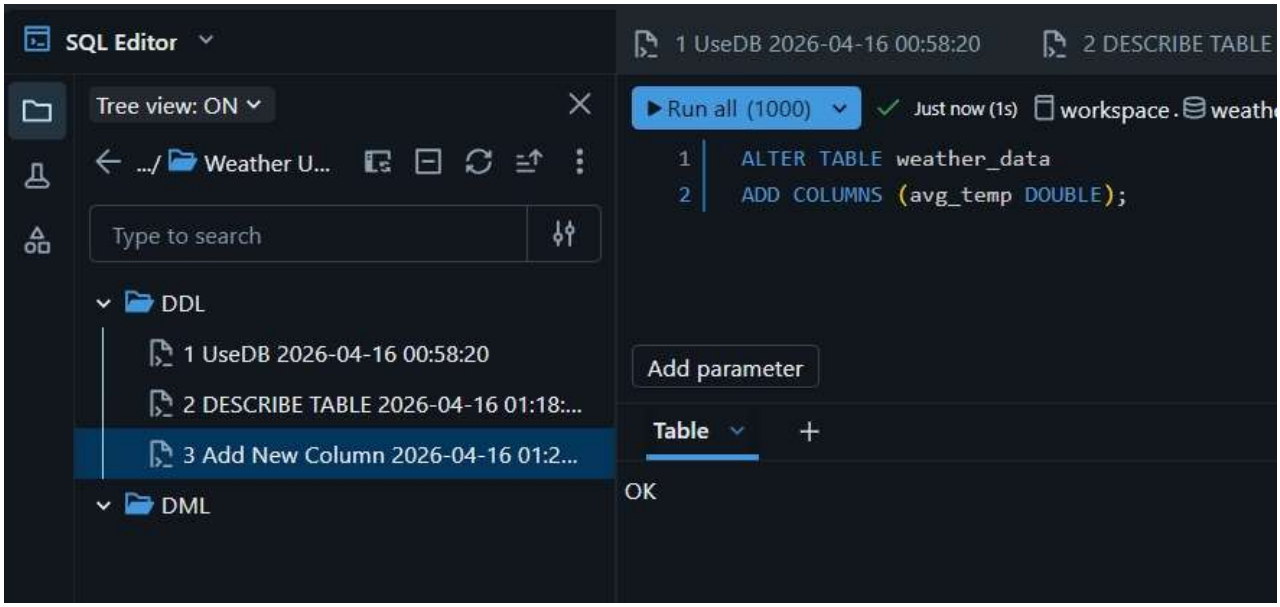
6.1 Use Database



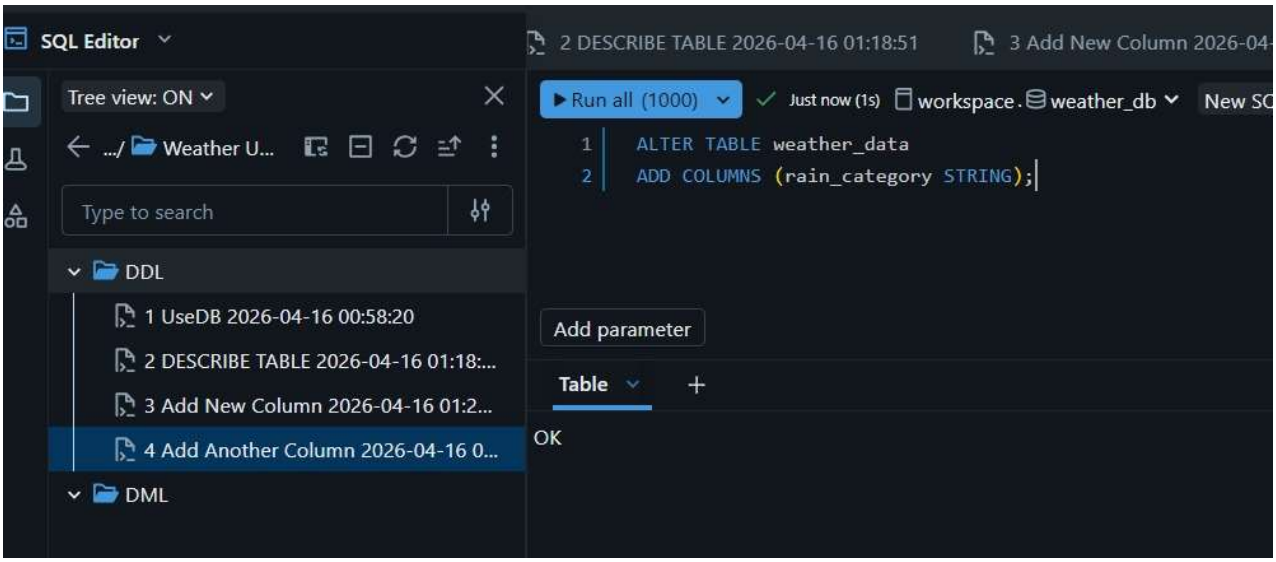
6.2 Describe Table



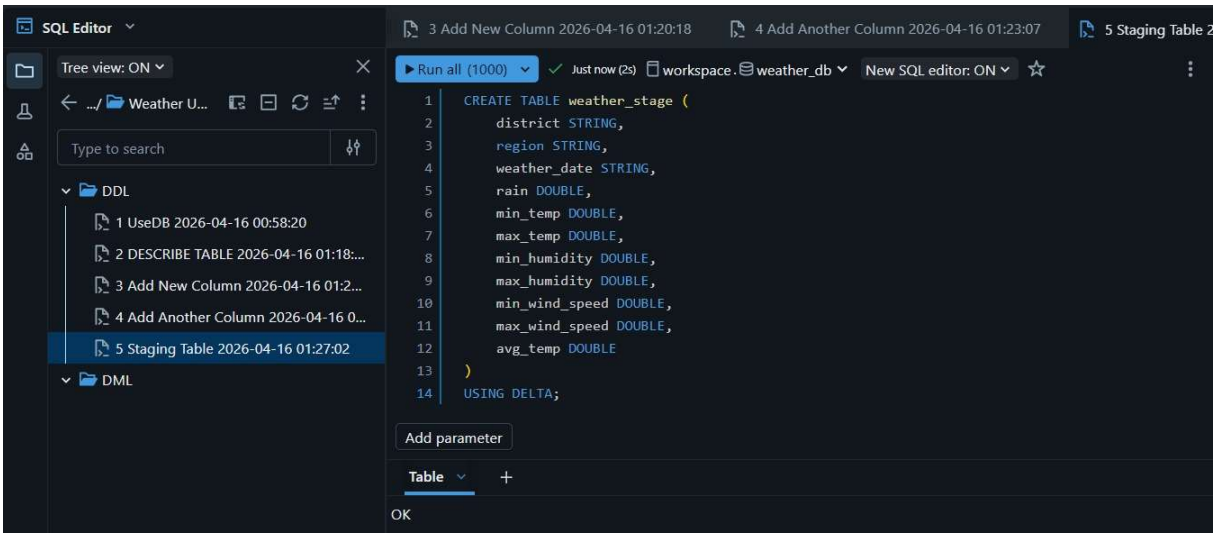
6.3 Add Column (avg_temp)



6.4 Add Column (rain_category)

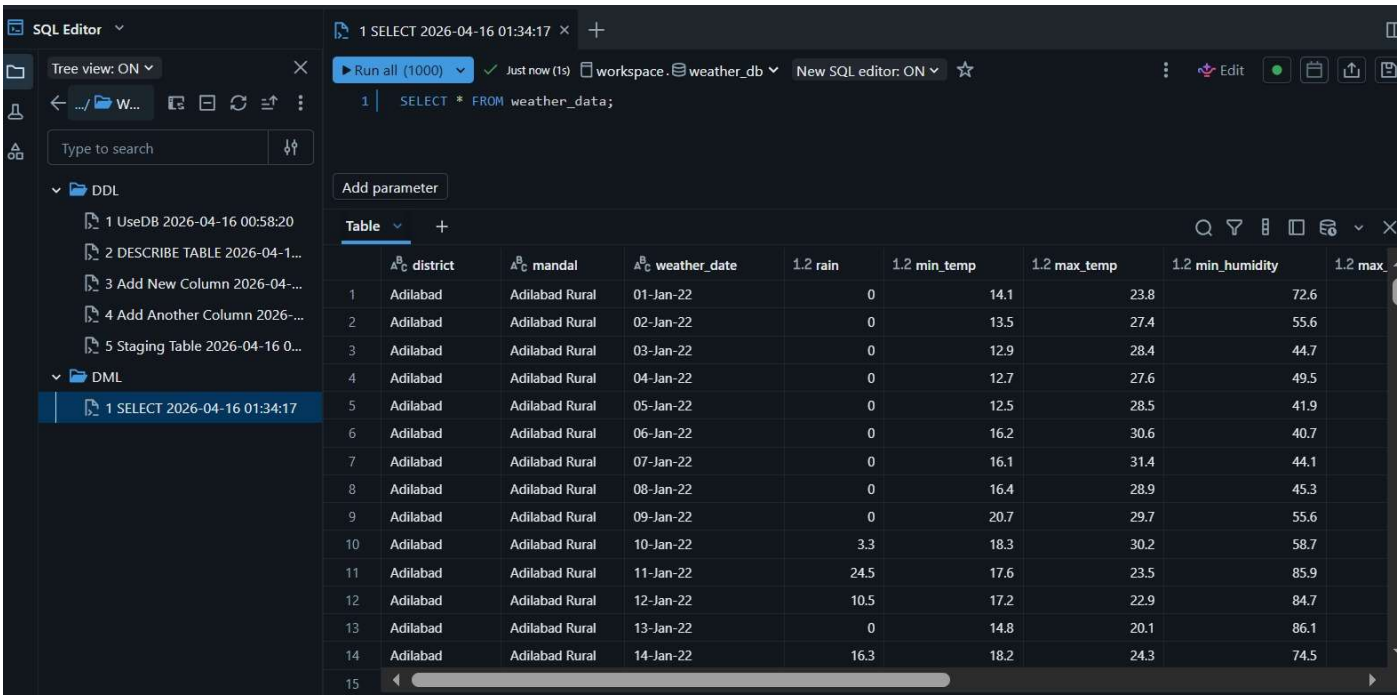


6.5 Create Staging Table



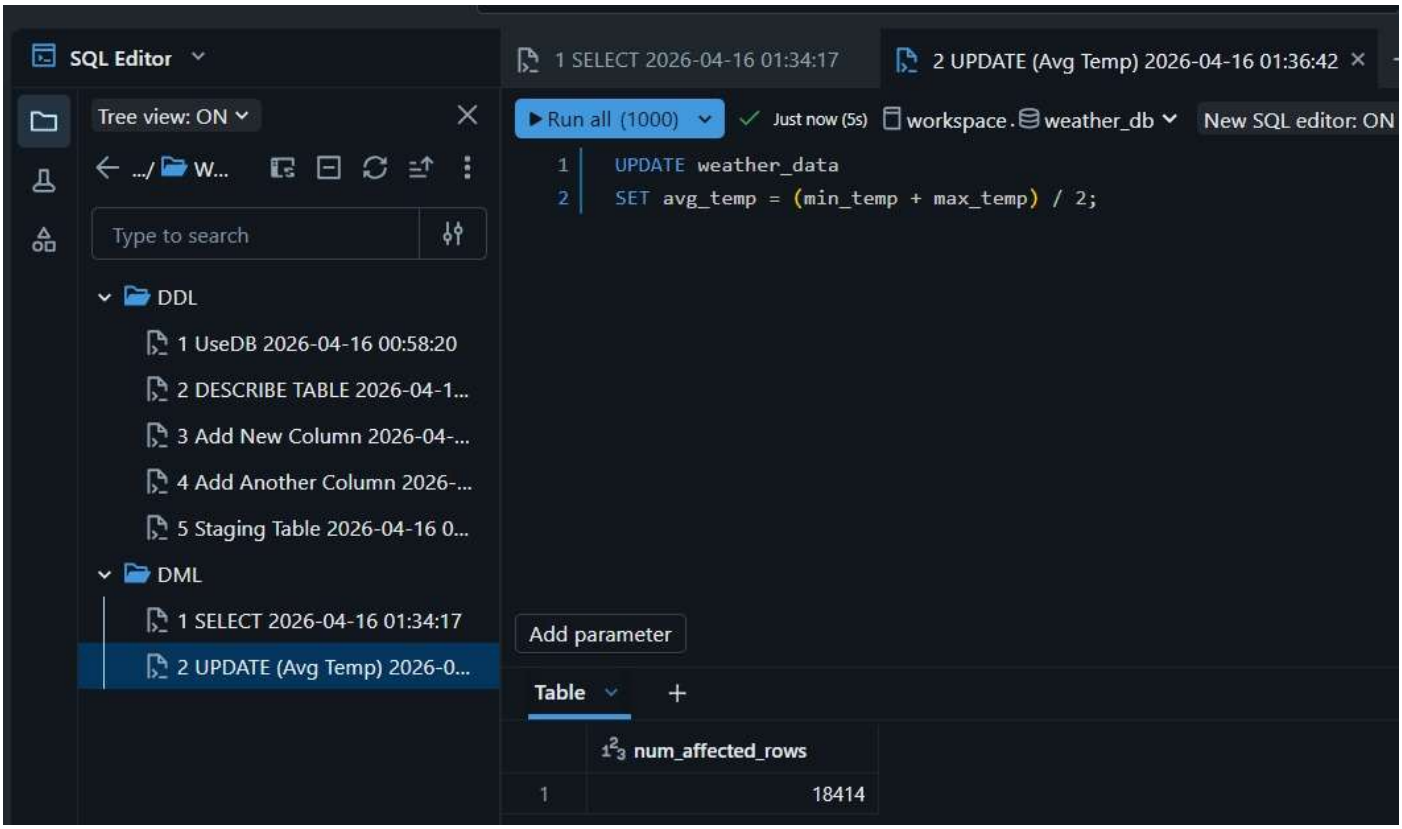
7. DML Operations

7.1 SELECT



	district	mandal	weather_date	rain	min_temp	max_temp	min_humidity	max_
1	Adilabad	Adilabad Rural	01-Jan-22	0	14.1	23.8	72.6	
2	Adilabad	Adilabad Rural	02-Jan-22	0	13.5	27.4	55.6	
3	Adilabad	Adilabad Rural	03-Jan-22	0	12.9	28.4	44.7	
4	Adilabad	Adilabad Rural	04-Jan-22	0	12.7	27.6	49.5	
5	Adilabad	Adilabad Rural	05-Jan-22	0	12.5	28.5	41.9	
6	Adilabad	Adilabad Rural	06-Jan-22	0	16.2	30.6	40.7	
7	Adilabad	Adilabad Rural	07-Jan-22	0	16.1	31.4	44.1	
8	Adilabad	Adilabad Rural	08-Jan-22	0	16.4	28.9	45.3	
9	Adilabad	Adilabad Rural	09-Jan-22	0	20.7	29.7	55.6	
10	Adilabad	Adilabad Rural	10-Jan-22	3.3	18.3	30.2	58.7	
11	Adilabad	Adilabad Rural	11-Jan-22	24.5	17.6	23.5	85.9	
12	Adilabad	Adilabad Rural	12-Jan-22	10.5	17.2	22.9	84.7	
13	Adilabad	Adilabad Rural	13-Jan-22	0	14.8	20.1	86.1	
14	Adilabad	Adilabad Rural	14-Jan-22	16.3	18.2	24.3	74.5	

7.2 UPDATE (Average Temperature)



	num_affected_rows
1	18414

7.3 UPDATE (Rain Category)

```
UPDATE weather_data
SET rain_category =
  CASE
    WHEN rain = 0 THEN 'No Rain'
    WHEN rain < 10 THEN 'Low'
    WHEN rain < 50 THEN 'Moderate'
    ELSE 'Heavy'
  END;
```

	num_affected_rows
1	18414

7.4 INSERT

```
INSERT INTO weather_data VALUES
('TestDistrict','TestMandal','2026-04-15',5,20,30,50,90,2,10,25,'Low');
```

	num_affected_rows	num_inserted_rows
1	1	1

7.5 DELETE

SQL Editor

4 INSERT 2026-04-16 01:39:27 | 5 DELETE 2026-04-16 01:40

Tree view: ON

Run all (1000) | Just now (2s) | workspace . weather_db

```
DELETE FROM weather_data
WHERE min_temp < 0;
```

Type to search

- DDL
  - 1 UseDB 2026-04-16 00:58:20
  - 2 DESCRIBE TABLE 2026-04-1...
  - 3 Add New Column 2026-04-...
  - 4 Add Another Column 2026-...
  - 5 Staging Table 2026-04-16 0...
- DML
  - 1 SELECT 2026-04-16 01:34:17
  - 2 UPDATE (Avg Temp) 2026-0...
  - 3 UPDATE (Rain Category) 20...
  - 4 INSERT 2026-04-16 01:39:27
  - 5 DELETE 2026-04-16 01:40:51

Add parameter

Table

	num_affected_rows
1	0

7.6 Insert into Staging Table

SQL Editor

4 INSERT 2026-04-16 01:39:27 | 5 DELETE 2026-04-16 01:40:51 | 6 Insert into Stage 2026

Tree view: ON

Run all (1000) | Just now (2s) | workspace . weather_db | New SQL editor: ON

```
INSERT INTO weather_stage VALUES
('Adilabad','Adilabad Rural','01-Jan-22',12,15,25,60,90,1,5,20);
```

Type to search

- DDL
  - 1 UseDB 2026-04-16 00:58:20
  - 2 DESCRIBE TABLE 2026-04-1...
  - 3 Add New Column 2026-04-...
  - 4 Add Another Column 2026-...
  - 5 Staging Table 2026-04-16 0...
- DML
  - 1 SELECT 2026-04-16 01:34:17
  - 2 UPDATE (Avg Temp) 2026-0...
  - 3 UPDATE (Rain Category) 20...
  - 4 INSERT 2026-04-16 01:39:27
  - 5 DELETE 2026-04-16 01:40:51
  - 6 Insert into Stage 2026-04-1...

Add parameter

Table

	num_affected_rows	num_inserted_rows
1	1	1

7.7 MERGE (Upsert)

```
MERGE INTO weather_data t
USING weather_stage s
ON t.district = s.district
AND t.weather_date = s.weather_date

WHEN MATCHED THEN UPDATE SET
    t.rain = s.rain

WHEN NOT MATCHED THEN INSERT (
    district,
    mandal,
    weather_date,
    rain,
    min_temp,
```

	num_affected_rows	num_updated_rows	num_deleted_rows	num_inserted_rows
1	18	18	0	0

7.8 INSERT OVERWRITE

```
INSERT OVERWRITE weather_data
SELECT * FROM weather_data WHERE rain > 0;
```

	num_affected_rows	num_inserted_rows
1	1741	1741

8. Key Outcomes

Successfully implemented **DDL operations**:

- ✓ CREATE TABLE
- ✓ ALTER TABLE (ADD COLUMN)
- ✓ DESCRIBE

Successfully implemented **DML operations**:

- ✓ SELECT
- ✓ INSERT
- ✓ UPDATE
- ✓ DELETE
- ✓ MERGE
- ✓ INSERT OVERWRITE

9. Learning Outcomes

- Understanding of Delta Lake architecture
- Practical implementation of DDL and DML
- Handling real-world datasets in Databricks
- Data transformation and cleaning
- Schema modification and data manipulation

10. Challenges Faced

- Column naming issues due to spaces and special characters
- Schema mismatch during MERGE operation
- Delta Lake restrictions on column modification

11. Conclusion

This project demonstrated how Databricks can be used to efficiently manage and manipulate data using DDL and DML operations. The implementation provided hands-on experience with real-world data processing and highlighted the importance of schema design and data consistency.